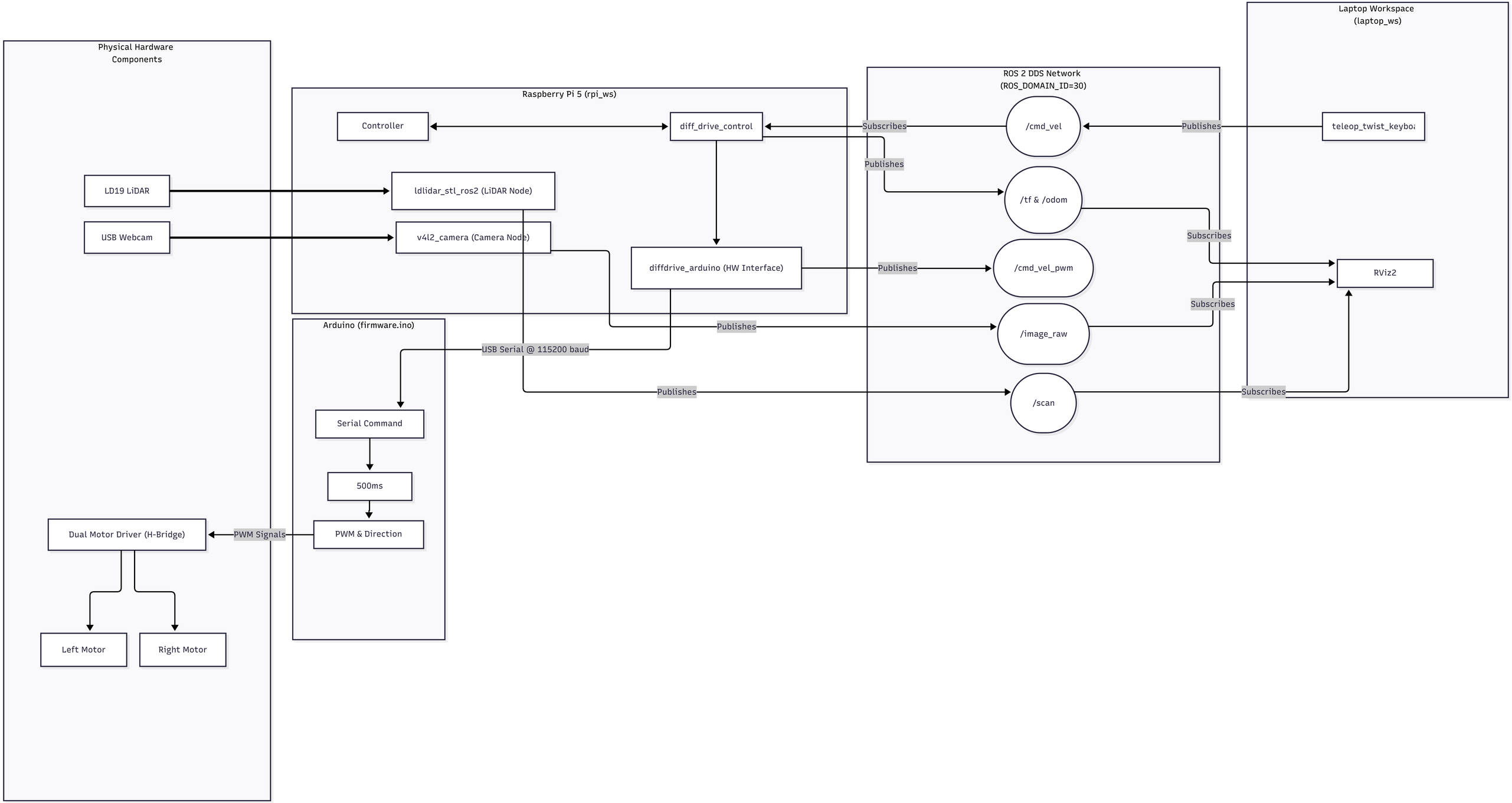
Physical Hardware Components
LD19 LiDAR
USB Webcam
Dual Motor Driver (H-Bridge)
Left Motor
Right Motor
Raspberry Pi 5 (rpi_ws)
Controller
diff_drive_control
ldlidar_stl_ros2 (LiDAR Node)
v4l2_camera (Camera Node)
diffdrive_arduino (HW Interface)
Arduino (firmware.ino)
Serial Command
500ms
PWM & Direction
PWM Signals
USB Serial @ 115200 baud
ROS 2 DDS Network
(ROS_DOMAIN_ID=30)
/cmd_vel
/tf & /odom
/cmd_vel_pwm
/image_raw
/scan
Subscribes
Publishes
Laptop Workspace
(laptop_ws)
teleop_twist_keyboa
RViz2